

## **Assignment No. 1 – Requirement Analysis and User Story Writing**

**Domain: QA (Quality Assurance)**

**Organization: 10Pearls Pakistan**

**Name: Zainab Arif**

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### **Project Overview:**

The client wants an Online Book Store which will enable users to browse, search, and buy books over the internet. It should provide a user-friendly interface, safe transactions, and account management features.

### **Refined User Requirements:**

User requirements emphasize the precise needs, expectations, and limitations of the end users or stakeholders using the software system.

#### **1) Core User Requirements (Needs)**

##### **User-Friendly Interface:**

An intuitive website or app which is simple to use, making it easy for users to find what they're looking for without difficulty.

##### **Personalized Recommendations:**

Book recommendations specifically for users based on previous purchases, browsing history, and declared interests in order to stimulate further exploration and interaction.

##### **Comprehensive Book Information:**

In-depth product pages with a plot summary, author biography, reviews, ratings, formats available, and sample chapters or previews.

##### **Efficient Search and Filtering:**

A robust search function with filters by genre, author, price, publication date, and other important criteria.

##### **Secure and Streamlined Checkout:**

An easy, secure checkout process with multiple payment methods and explicit order confirmation.

#### **2) User Preferences (Likes)**

##### **Format Flexibility:**

Options for physical books, e-books, and audiobooks to cater to various reading habits and accessibility needs.

##### **Loyalty Programs and Discounts:**

Programs that reward repeat customers with discounts and special offers.

**Community and Social Features:**

Features like book clubs, user-submitted reviews, wishlists, and the ability to share reading progress with friends.

**Author and Publisher Insights:**

Access to interviews with authors, publisher news, and curated lists to deepen the engagement with the literary world.

**Pre-orders and New Releases:**

Early pre-order access to new titles and keep informed about releases.

**3) User Expectations (Assumptions)**

**Competitive Pricing:**

Reasonable and competitive prices for books, possibly with price-matching commitments.

**Fast and Reliable Delivery:**

Physical book delivery on time with definite shipping schedules and tracking facilities.

**Clear Return Policies:**

Easy and transparent return policy for damaged or unsatisfactory books.

**Responsive Customer Service:**

Easily accessible and helpful customer support to promptly resolve issues via multiple channels such as chat, email, or phone.

**Trust and Security:**

A safe platform that safeguards user information and provides secure transactions.

**Functional Requirements:**

Functional requirements specify what a system does, describing its features and capabilities.

**User Registration and Profiles:**

Users should be able to create and maintain accounts, securely log in, and access their personal data, order history, and stored addresses.

**Book Catalog and Search:**

The system must present a list of books with such details as title, author, genre, price, and cover image. Users should be able to search for books based on different criteria and use filters.

**Shopping Cart:**

Users should be able to add or remove books and modify the quantity of books in their shopping cart, and a dynamically updating total cost.

**Checkout and Payment:**

The site must be able to offer a secure, multi-step checkout process where customers can enter shipping information, choose shipping modes, and make payments via embedded gateways.

**Order Management:**

Users must be able to see their previous orders and current orders and monitor their status.

**Reviews and Ratings:**

The system should enable customers to leave reviews and rate the books they have bought.

**User Management:**

Administrators require the ability to control registered users, including giving them permission or removing their accounts.

**Inventory Management:**

The system must manage book stocks, maintaining stock levels when books are sold and being able to restock.

**Order Processing:**

Admins should be able to process orders, control shipping, and accept returns or cancellations.

**Content Management:**

The ability to add new books into the catalog, edit existing book information, and maintain categories is vital for the administrator.

**Non Functional Requirements:**

Non-functional requirements are quality attributes that ensure the system meets user expectations and can work effectively under various conditions.

**Performance:**

Specifies the responsiveness of the system.

Example: Pages must load in three seconds under normal user load.

**Security:**

Is concerned with the protection of user data as well as financial transactions.

Example: User transactions and data should be encrypted using SSL/TLS.

**Usability:**

Focuses on how easy and intuitive the platform is for users to use.

Example: The user interface should be user-friendly and intuitive.

**Scalability:**

The capacity of the system to expand and handle rising user numbers and transactions over time.

Example: The architecture of the platform needs to be scalable enough to handle a growing user base and increased inventory without performance degradation.

**Reliability:**

Sees to it that the system is reliable and available.

Example: The system must have a high availability level (e.g., 99.9% up-time).

**Compatibility:**

Refers to how the system can execute on numerous hardware platforms, operating systems, and browsers.

Example: The site must function correctly on major web browsers such as Chrome, Firefox, and Safari.

**Maintainability:**

How simple it is to modify, fix bugs, and enhance the system in the future.

Example: The developers must be able to deploy updates or new functionality with zero or minimal downtime for the users.

**Portability:**

The ability to run the system on different environments or platforms.

Example: The site must work on different devices such as PCs, tablets, and smartphones.

**Functional Acceptance Criteria:**

**User Registration & Login:**

The application permits new users to register with a unique email address and password.

Users can log in using their registered credentials.

Users can update their account information (e.g., address, password).

**Book Catalog and Search:**

Users can browse books by predefined categories.

Results must be presented in less than two seconds.

No results must display a clear "No books found" message.

Users can search for books based on title, author, or other relevant information.

Search results must be unambiguous, so users can pick a book of interest.

**Shopping Cart Management:**

User can add books to cart.

Users can edit quantities or delete items.

The cart must dynamically display the total price at all times.

**Checkout and Payment:**

Users can checkout from their shopping cart.

The system needs to support a safe way of making payments (e.g., through credit card).

Users should get a confirmation on successful completion of an order.

**Order History:**

Users can view their past orders and order history.

**Notifications:**

Notifications (e.g., by SMS) on the status of their order should be sent to users.

**Non Functional Acceptance Criteria:****Security:**

Secure protocols such as SSL should be used to safeguard sensitive user and payment data.

All system activity should be logged.

**User-Friendliness:**

The interface must be easy to use and interactive, assisting customers in making decisions.

**Reliability:**

The system must be operational 24/7.

It must be able to accommodate peak usage periods without suffering from performance losses.

**Data Accuracy:**

The system should keep accurate records of customers, books, sales, and deliveries.

**Hardware/Software Compatibility:**

The system should be compatible with common web browsers such as Chrome and Firefox.

**User Stories (Gherkin Format)****Feature 1 – User Registration**

As a new visitor

I want to register with my email and password

So that I can create a personal account.

Gherkin:

Given I am on the registration page

When I enter a valid email and password

And I click the "Register" button

Then my account should be created successfully

And I should receive a confirmation message

**Feature 2 – User Login**

As a registered user

I want to log in using my credentials

So that I can view my account and order history.

Gherkin:

Given that I have a registered account

When I put in my proper email and password

And I press the "Login" button  
Then I should be logged into my account  
And I should view my dashboard

### Feature 3 – Book Search

As a customer  
I want to search for books by title, author, or genre  
So that I can locate the books that I desire efficiently.  
Gherkin:

Given that I am on the search bar  
When I input a valid title of a book  
And I click the "Search" button  
Then the system should provide matching results  
And results should be shown within 2 seconds

### Feature 4 – Book Details

As a customer  
I want to see detailed information about a book  
So that I may make an informed decision of purchasing.  
Gherkin:

Given that I am perusing the catalog of books  
When I click on one particular book  
Then I should be able to view information such as title, author, cost, rating, and reviews  
And I should be able to view a sample chapter or preview

### Feature 5 – Shopping Cart Management

As a customer  
I want to add, edit, or delete books from my cart  
So that I can organize my purchases prior to checkout.  
Gherkin:

Given that I have chosen a book  
When I put it in the shopping cart  
Then the cart must show the book that is chosen  
And the total amount must reflect dynamically

### Feature 6 – Checkout & Payment

As a customer  
I want to checkout and pay securely for my books  
So that I am able to finalize my purchase.  
Gherkin:  
Given I have products in my shopping cart  
When I click on checkout  
And I provide correct shipping and payment information

Then the system must process my order

And I must get an order confirmation

### Feature 7 – Order History & Tracking

As a customer

I want to be able to see my order history and track order status

So that I can be informed about my purchases.

Gherkin:

Given that I am logged into my account

When I visit "My Orders"

Then I should be able to see a list of my previous and current orders

And I should be able to track the status of in-progress orders

### Sprint Planning:

To provide systematic delivery of the Online Book Store project, the work has been broken down into three sprints. Each sprint is intended to provide coverage of certain features and functionalities, along with ensuring incremental delivery of value to the end users.

#### Sprint 1 – User Module (Week 1)

Scope:

User Registration and Login

Book Catalog and Search

Shopping Cart Management

Deliverables:

By the end of Sprint 1, the system will enable users to log in securely, browse the catalog, create accounts, and manage shopping carts by adding or deleting books. This sprint centers on the development of user interaction with the platform as its foundation.

#### Sprint 2 – Transaction Module (Week 2)

Scope:

Checkout and Payment

Order History

Notifications

Deliverables:

By Sprint 2 end, users will be able to safely make payments, get confirmations, and view order history. Notifications (order confirmation, shipping updates, etc.) will also be made. This sprint makes sure that users are able to successfully transact on the platform.

#### Sprint 3 – Admin & System Module (Week 3)

Scope:

User Management (Admin)

Inventory and Order Processing

Non-Functional Requirements (Performance, Security, Reliability)

Deliverables:

By Sprint 3, the admin will have user account management, inventory update, and order processing capabilities. Also, non-functional requirements like security (data protection, SSL), performance (response speed), and reliability will be implemented. This sprint makes sure that the system is stable, secure, and ready for deployment in real-world environments.

### Sprint Summary Table

| Sprint                           | Duration | Scope                                                   | Key Deliverables                                                     |
|----------------------------------|----------|---------------------------------------------------------|----------------------------------------------------------------------|
| Sprint 1 – User Module           | Week 1   | Registration, Login, Catalog, Cart                      | Users can register, browse, and manage carts                         |
| Sprint 2 – Transaction Module    | Week 2   | Checkout, Payment, Order History, Notifications         | Users can securely purchase and view order history                   |
| Sprint 3 – Admin & System Module | Week 3   | User Management, Inventory, Non-Functional Requirements | Admin can manage data; system meets security & performance standards |

### Conclusion:

This user story writing and requirement analysis exercise clearly identified:

Functional & Non-functional requirements for the Online Book Store

User stories in Gherkin format to record customer interactions

Acceptance criteria to ensure functionality and quality

Sprint planning to break down development into smaller iterations

Following this systematic plan, the Online Book Store can be built using a user-centric approach with usability, security, and scalability.